Causal discovery for robust predictions

Homer Durand

Image Processing Lab (ISP group)

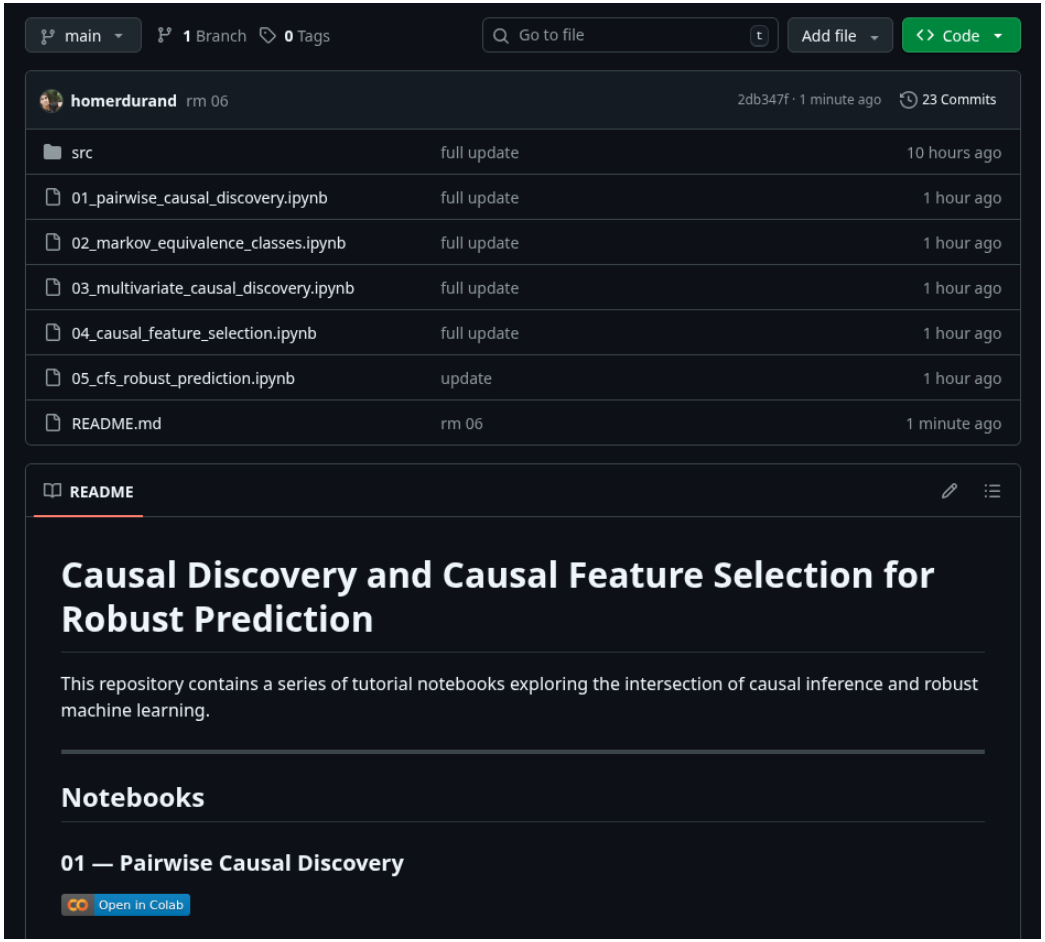
Mar 2026

0 Outline

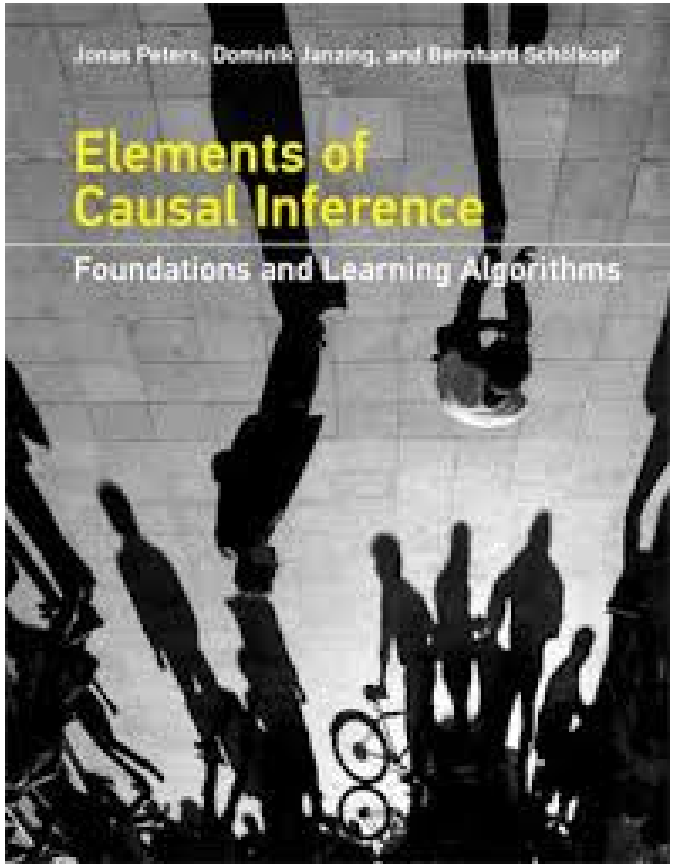
1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

0 Github repo

github.com/WinterSchool2026/zh03-causal-discovery-robust-predictions



0 Literature - Statistical Causality



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Elements of Causal Inference (2017)

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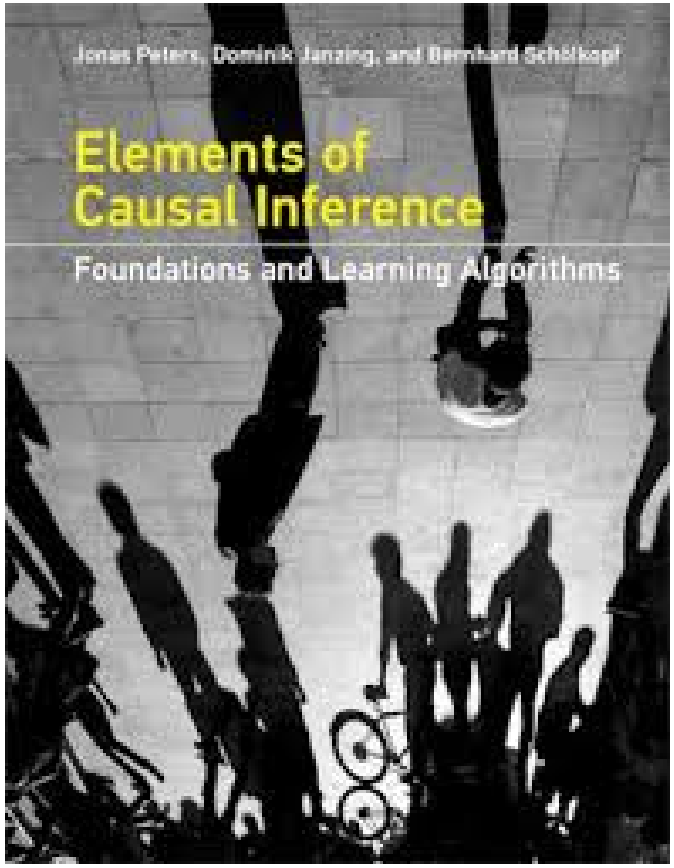

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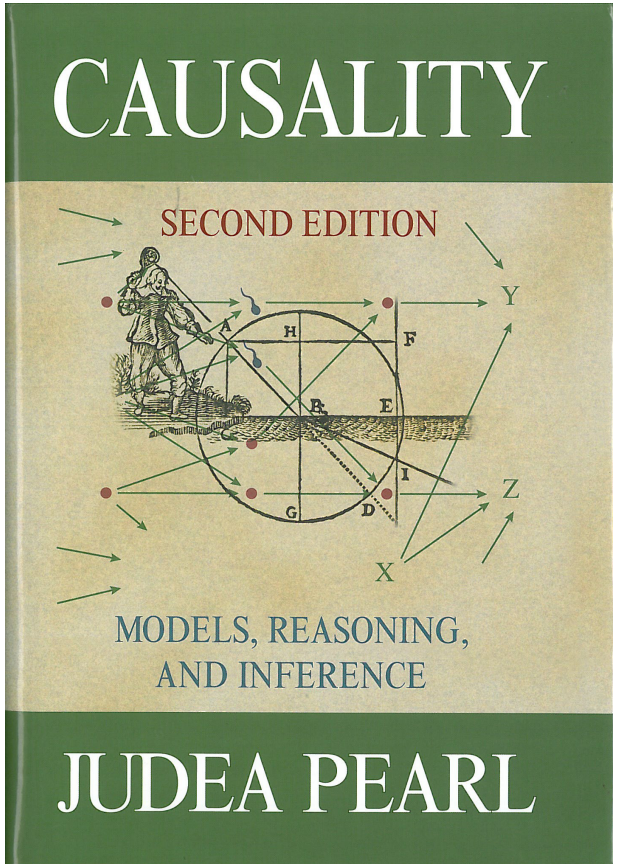

Judea Pearl
Causality (2009)

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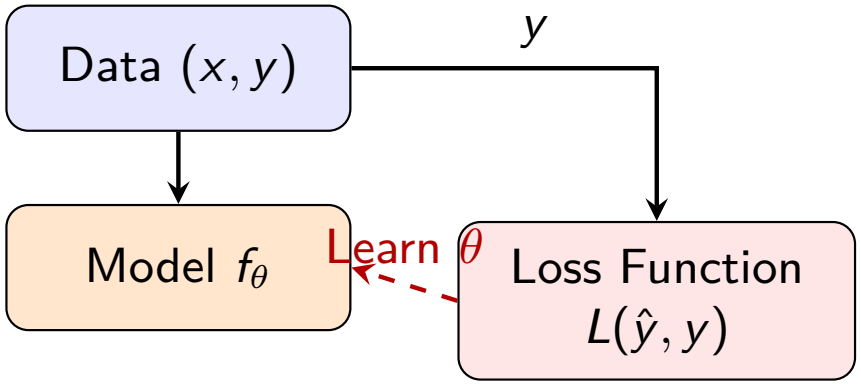

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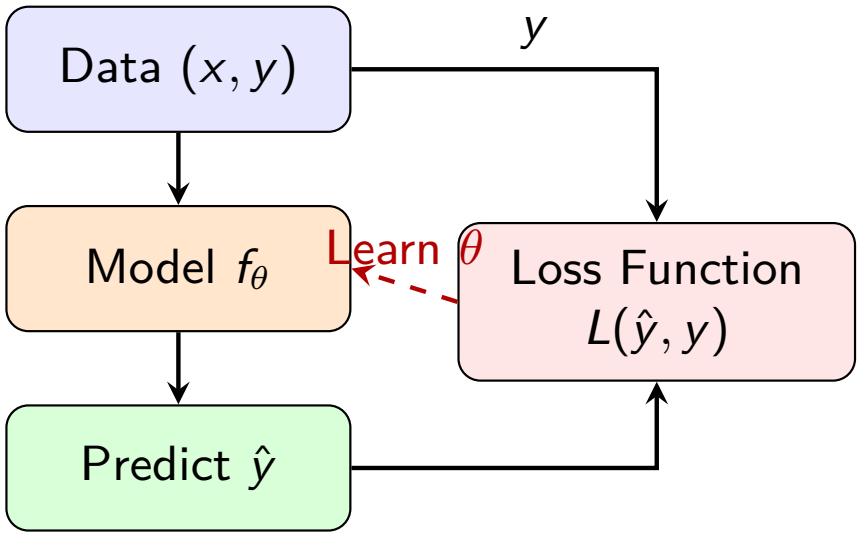

1 Machine Learning 101

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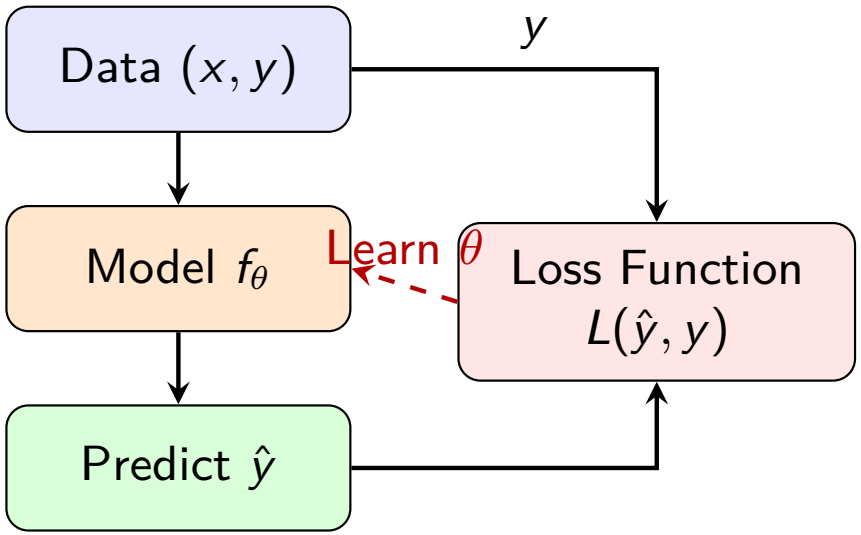
Example:

- x = satellite observations (SST, humidity, ...)
- y = precipitations
- f_θ = Convolutional Neural Network
- L = Mean Squared Error



1 Empirical Risk Minimisation (ERM)

ERM: find parameters $\hat{\theta}$ minimising average prediction error over training data:

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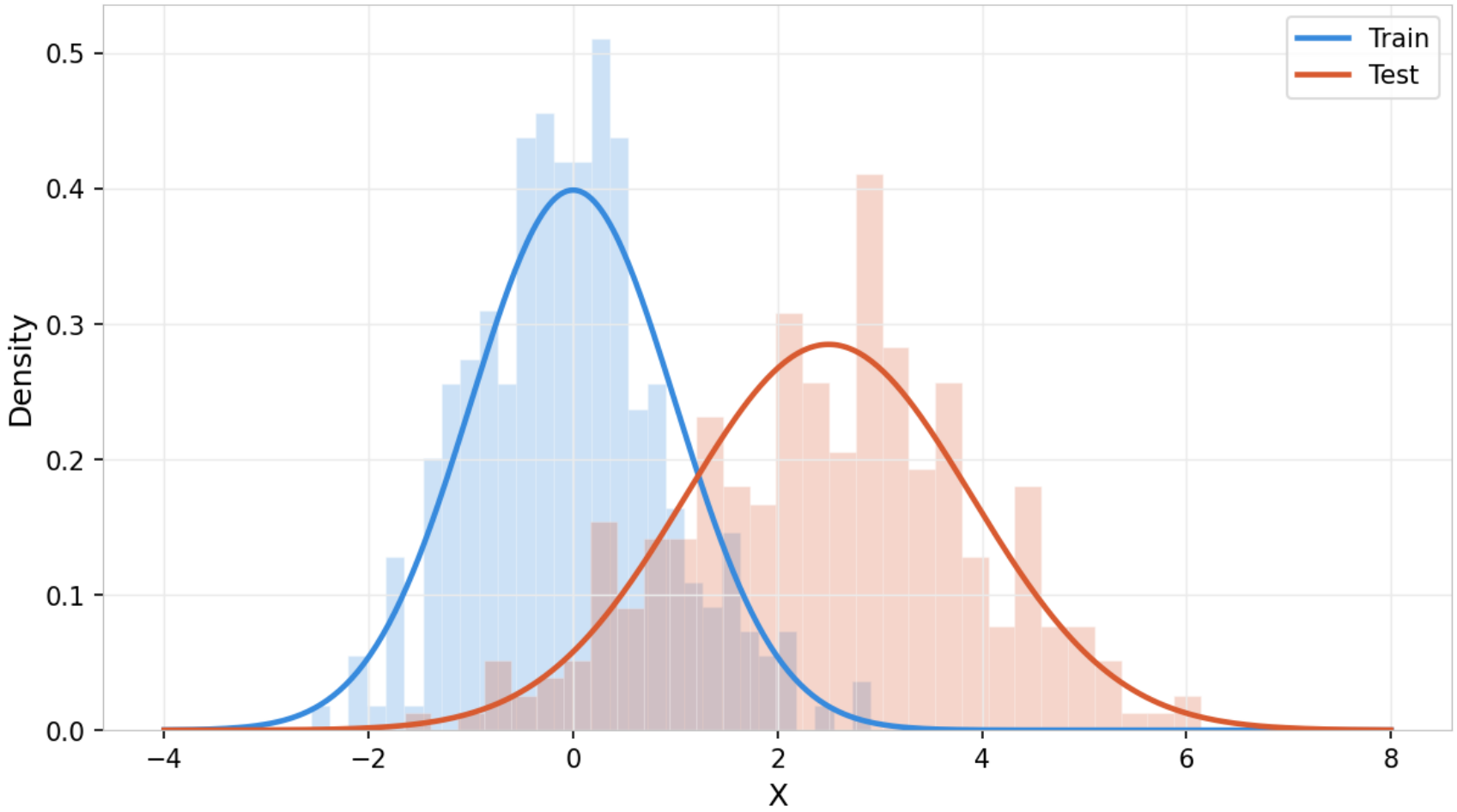
Optimality of ERM

Assuming i.i.d data, ERM is optimal: it minimizes the expected risk.

1 Distribution shifts



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Tipping points
AMOC slowdown, permafrost feedback: physical *mechanisms* themselves change

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- What makes some feature more robust?

  $\Rightarrow$ **Causal structure!**

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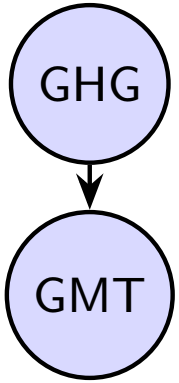

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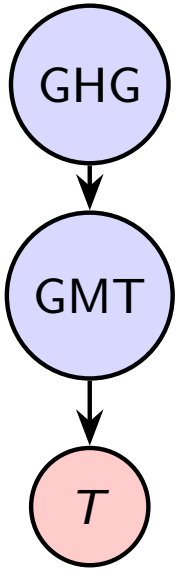

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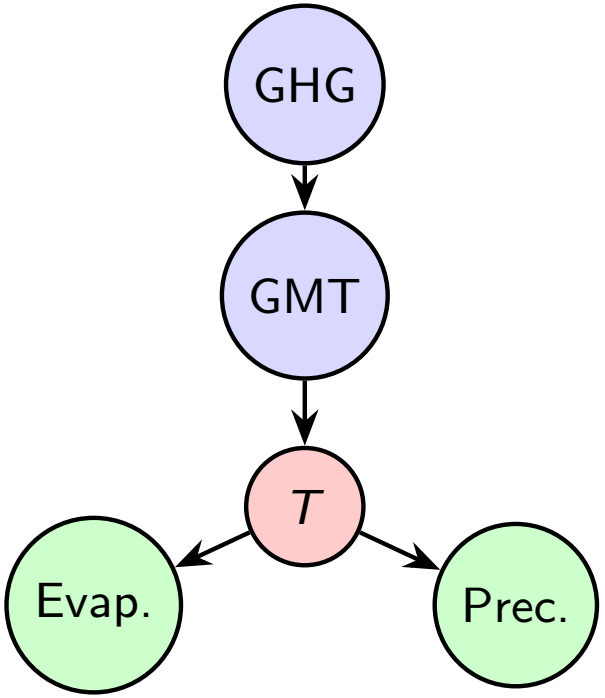

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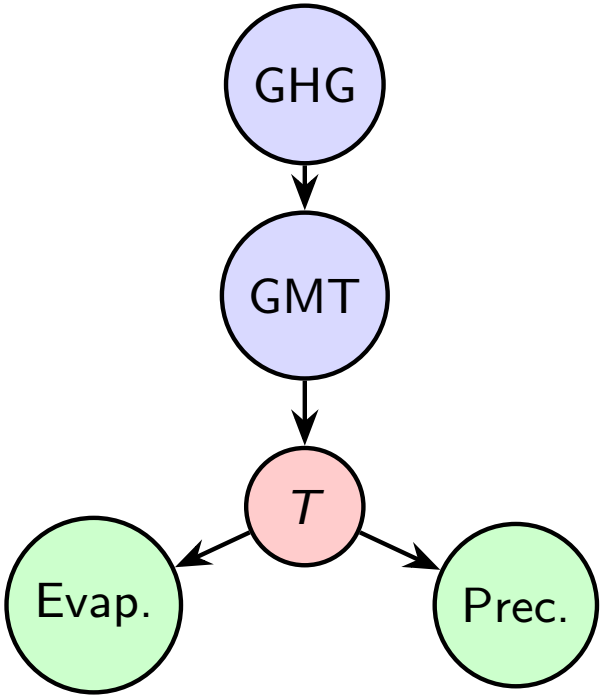

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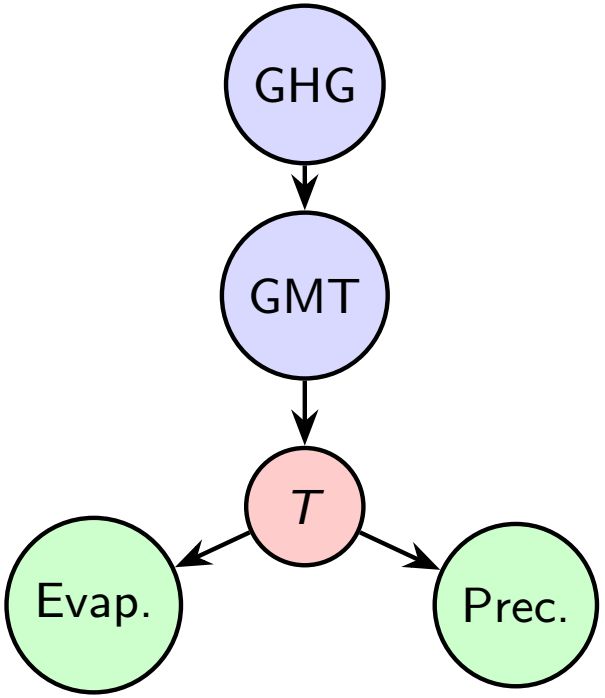

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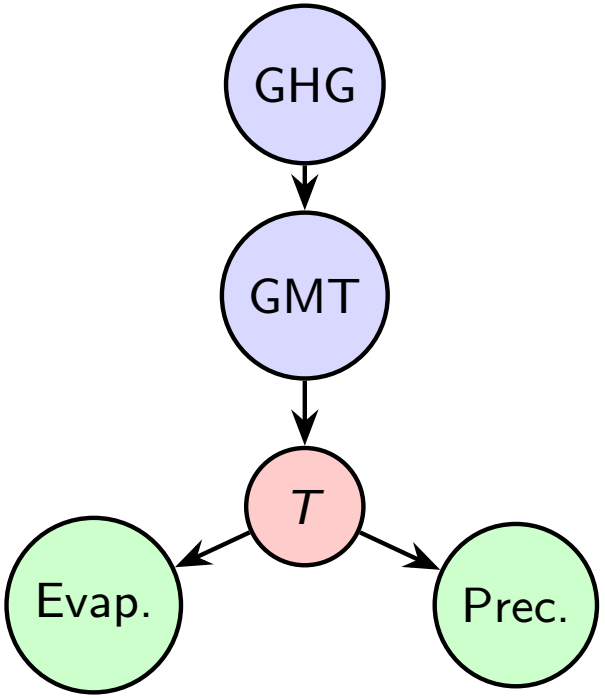

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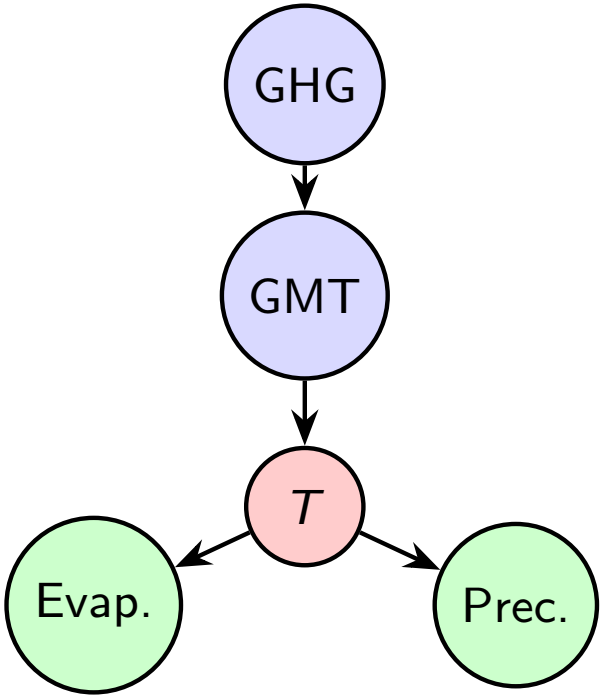
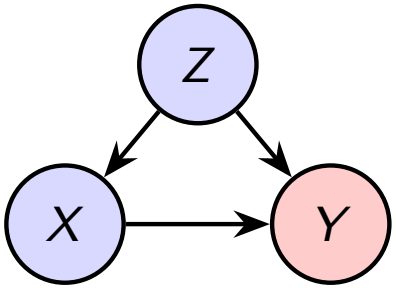


An intervention $do(X = x)$ sets a variable to a fixed value, overriding its natural causes, i.e. its **parents** in the graph.

Observational world



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2 Interventions: Cutting Arrows in the Graph

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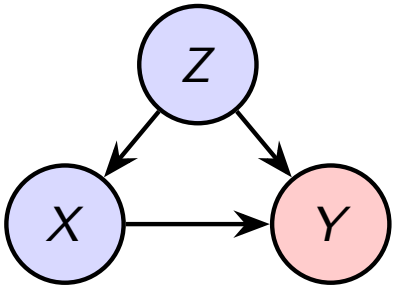

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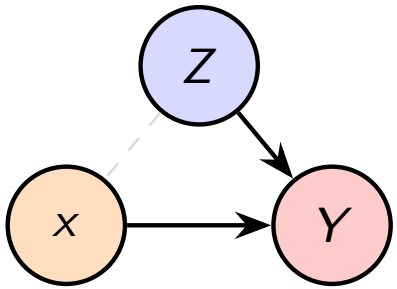
Manipulated world $do(X = x)$



arrow from Z is deleted

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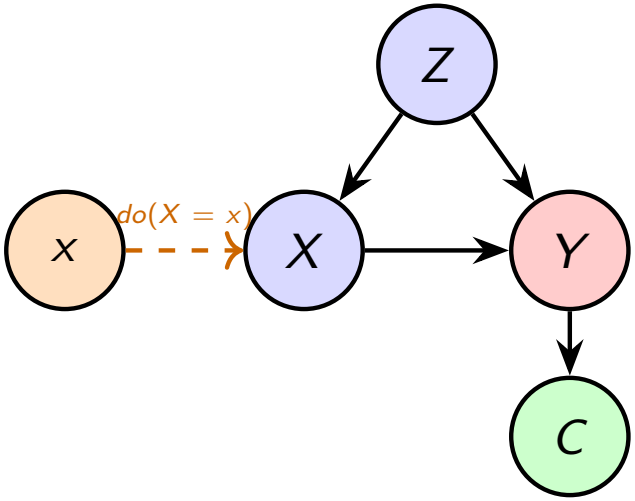

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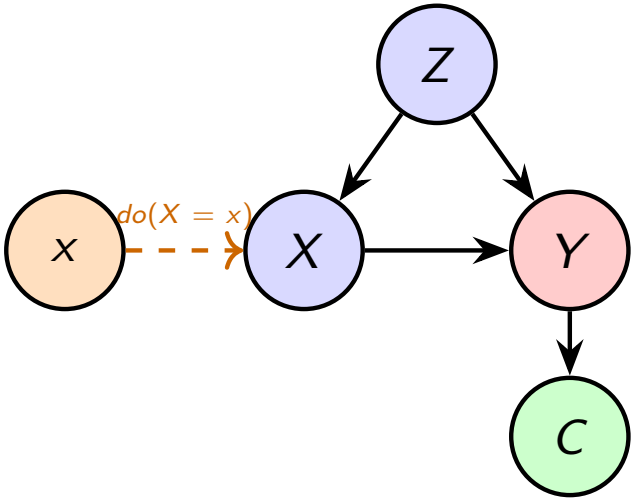
Intervention class $\mathcal{P}(\mathcal{G})$:

Fix a graph $\mathcal{G}$. The collection $\mathcal{P}(\mathcal{G}) = \{P^{do(\mathbf{X}_I=\mathbf{x})} : I \subseteq V, \mathbf{x}\}$ is the **interventional family** of $\mathcal{G}$.



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Assume the causal graph $\mathcal{G}$ is known. We ask: **which feature set** to use for predicting Y?

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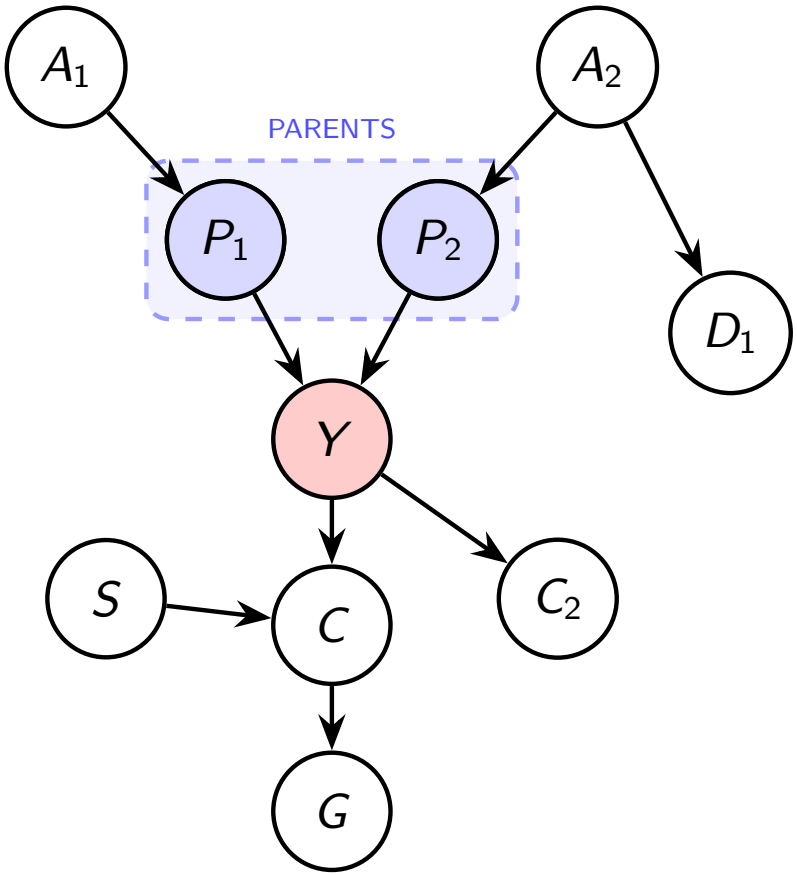

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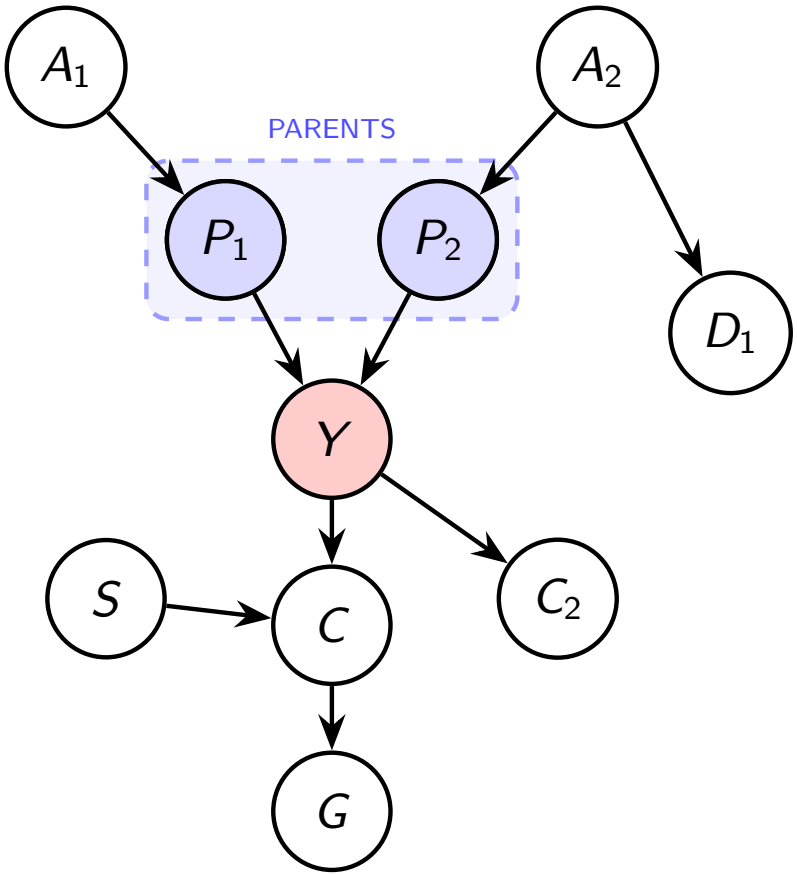

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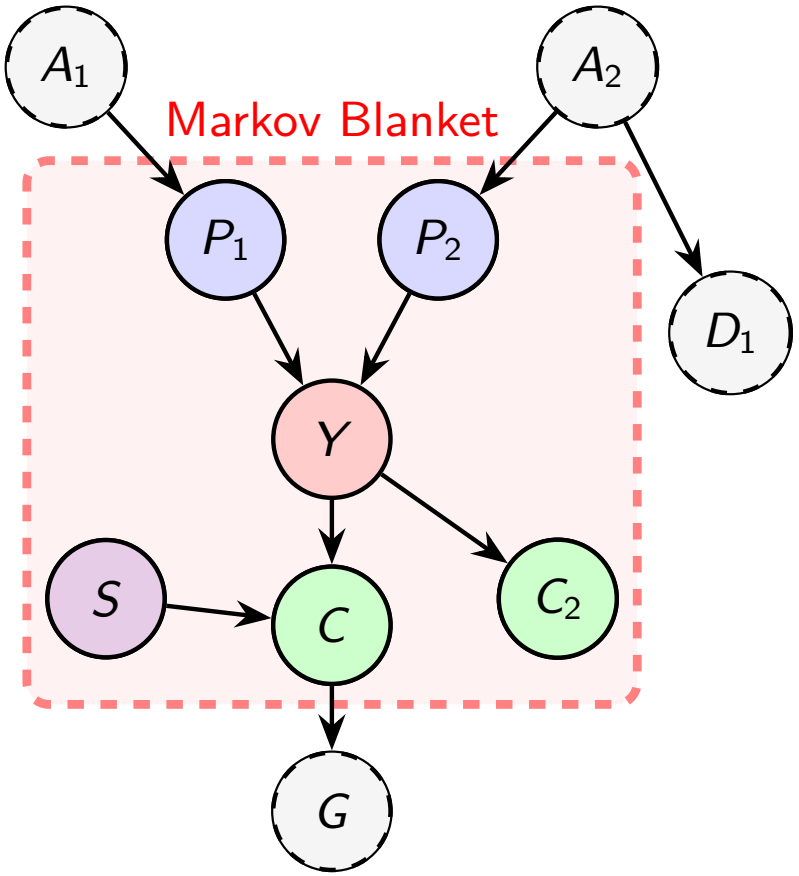

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No intervention (i.i.d data)	Lower	Higher
Intervention on *children*	✓	×
Intervention *non-MB* variables	✓	✓
Feature set size	Smaller	Larger
Learning from data	Harder	Easier

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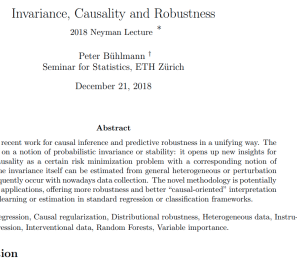
- **The right choice depends on knowing *which* variables will be intervened on.**



Invariance, Causality and Robustness

2018 Neyman Lecture *

Peter Bühlmann †
Seminar for Statistics, ETH Zürich

December 21, 2018

Abstract

We discuss recent work for causal inference and predictive robustness in a unifying way. The key idea relies on a notion of probabilistic invariance or stability: it opens up new insights for formulating causality as a certain risk minimization problem with a corresponding notion of robustness. The invariance itself can be estimated from general heterogeneous or perturbation data which frequently occur with nowadays data collection. The novel methodology is potentially useful in many applications, offering more robustness and better "causal-oriented" interpretation than machine learning or estimation in standard regression or classification frameworks.

Keywords: Anchor regression, Causal regularization, Distributional robustness, Heterogeneous data, Instrumental variables regression, Interventional data, Random Forests, Variable importance.

1 Introduction

ROYAL STATISTICAL SOCIETY
DATA | EVIDENCE | DECISIONS
Journal of the Royal Statistical Society
Statistical Methodology
Series B

J. R. Statist. Soc. B (2016)
78, *Part* 5, *pp.* 947–1012

Causal inference by using invariant prediction: identification and confidence intervals

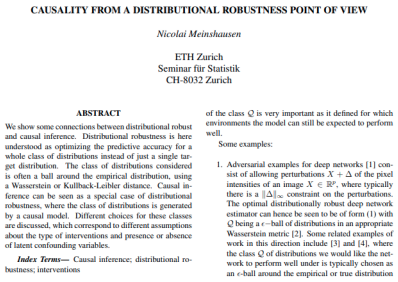
Jonas Peters

Max Planck Institute for Intelligent Systems, Tübingen, Germany, and Eidgenössiche Technische Hochschule Zürich, Switzerland

and Peter Bühlmann and Nicolai Meinshausen

Eidgenössiche Technische Hochschule Zürich, Switzerland

[*Read before* The Royal Statistical Society *at a meeting organized by the* Research Section *on Wednesday, May 11th, 2016,* Professor C. Leng *in the Chair*]



CAUSALITY FROM A DISTRIBUTIONAL ROBUSTNESS POINT OF VIEW

Nicolai Meinshausen

ETH Zurich
Seminar für Statistik
CH-8032 Zurich

ABSTRACT

We show some connections between distributional robust and causal inference. Distributional robustness is here understood as optimizing the predictive accuracy for a whole class of distributions instead of just a single target distribution. The class of distributions considered is often a ball around the empirical distribution, using a Wasserstein or Kullback-Leibler distance. Causal inference can be seen as a special case of distributional robustness, where the class of distributions is generated by a causal model. Different choices for these classes are discussed, which correspond to different assumptions about the type of interventions and presence or absence of latent confounding variables.

***Index Terms*—** Causal inference; distributional robustness; interventions

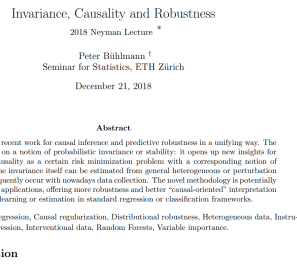
of the class $\mathcal{Q}$ is very important as it defined for which environments the model can still be expected to perform well.

Some examples:

1. Adversarial examples for deep networks [1] consist of allowing perturbations $X + \Delta$ of the pixel intensities of an image $X \in \mathbb{R}^p$, where typically there is a $\|\Delta\|_\infty$ constraint on the perturbations. The optimal distributionally robust deep network estimator can hence be seen to be of form (1) with $\mathcal{Q}$ being a $\epsilon-$ball of distributions in an appropriate Wasserstein metric [2]. Some related examples of work in this direction include [3] and [4], where the class $\mathcal{Q}$ of distributions we would like the network to perform well under is typically chosen as an ϵ-ball around the empirical or true distribution

Important Papers

- **Invariance, Causality and Robustness**
  Meinshausen (2018)



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We discuss recent work for causal inference and predictive robustness in a unifying way. The key idea relies on a notion of probabilistic invariance or stability: it opens up new insights for formulating causality as a certain risk minimization problem with a corresponding notion of robustness. The invariance itself can be estimated from general heterogeneous or perturbation data which frequently occur with nowadays data collection. The novel methodology is potentially useful in many applications, offering more robustness and better "causal-oriented" interpretation than machine learning or estimation in standard regression or classification frameworks.

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1 Introduction

ROYAL STATISTICAL SOCIETY
DATA | EVIDENCE | DECISIONS
Journal of the Royal Statistical Society
Statistical Methodology
Series B

J. R. Statist. Soc. B (2016)
78, *Part* 5, *pp.* 947–1012

Causal inference by using invariant prediction: identification and confidence intervals

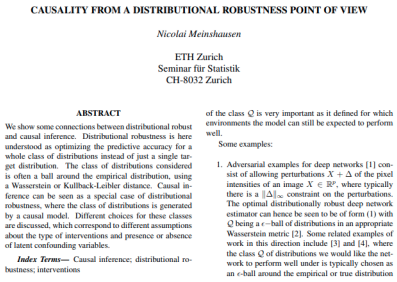
Jonas Peters

Max Planck Institute for Intelligent Systems, Tübingen, Germany, and Eidgenössiche Technische Hochschule Zürich, Switzerland

and Peter Bühlmann and Nicolai Meinshausen

Eidgenössiche Technische Hochschule Zürich, Switzerland

[*Read before* The Royal Statistical Society *at a meeting organized by the* Research Section *on Wednesday, May 11th, 2016,* Professor C. Leng *in the Chair*]



CAUSALITY FROM A DISTRIBUTIONAL ROBUSTNESS POINT OF VIEW

Nicolai Meinshausen

ETH Zurich
Seminar für Statistik
CH-8032 Zurich

ABSTRACT

We show some connections between distributional robust and causal inference. Distributional robustness is here understood as optimizing the predictive accuracy for a whole class of distributions instead of just a single target distribution. The class of distributions considered is often a ball around the empirical distribution, using a Wasserstein or Kullback-Leibler distance. Causal inference can be seen as a special case of distributional robustness, where the class of distributions is generated by a causal model. Different choices for these classes are discussed, which correspond to different assumptions about the type of interventions and presence or absence of latent confounding variables.

***Index Terms*—** Causal inference; distributional robustness; interventions

of the class $\mathcal{Q}$ is very important as it defined for which environments the model can still be expected to perform well.

Some examples:

1. Adversarial examples for deep networks [1] consist of allowing perturbations $X + \Delta$ of the pixel intensities of an image $X \in \mathbb{R}^p$, where typically there is a $\|\Delta\|_\infty$ constraint on the perturbations. The optimal distributionally robust deep network estimator can hence be seen to be of form (1) with $\mathcal{Q}$ being a $\epsilon-$ball of distributions in an appropriate Wasserstein metric [2]. Some related examples of work in this direction include [3] and [4], where the class $\mathcal{Q}$ of distributions we would like the network to perform well under is typically chosen as an ϵ-ball around the empirical or true distribution

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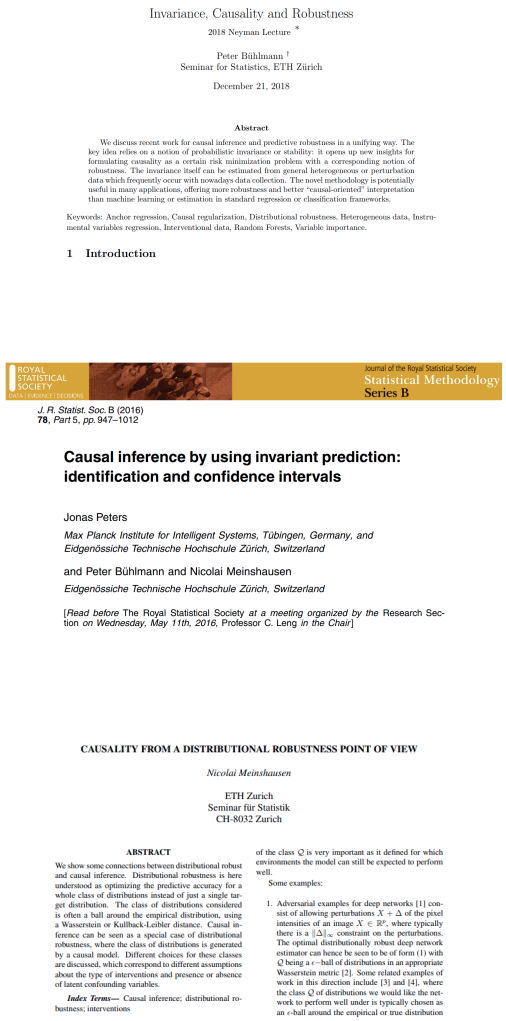

Invariance, Causality and Robustness

2018 Neyman Lecture *

Peter Bühlmann †
Seminar for Statistics, ETH Zürich

December 21, 2018

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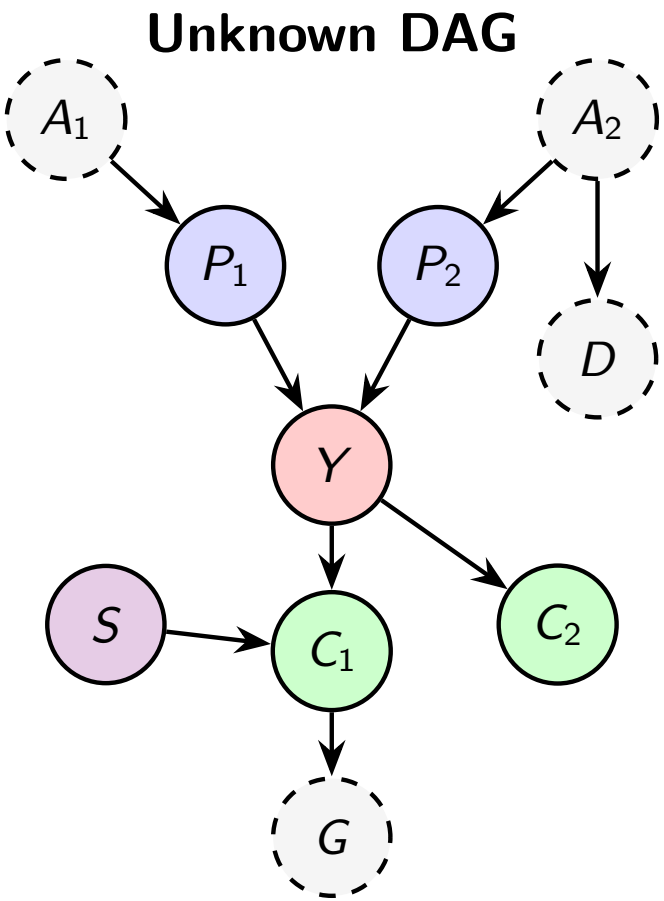
  Bühlmann (2020)

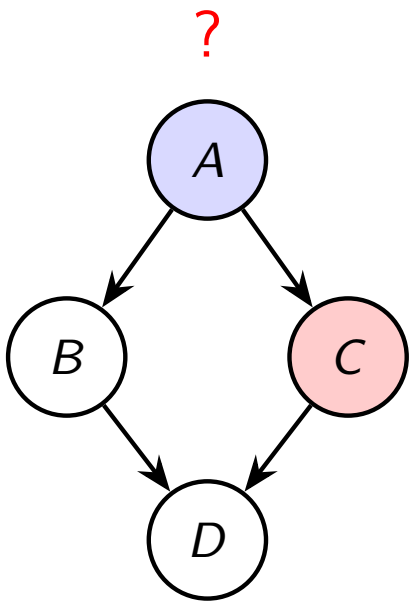
3 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion



	A_1	A_2	P_1	P_2	C_1	C_2	S	G	D	Y
0	0.8	-0.2	1.1	0.5	-0.7	0.3	-1.1	0.4	0.9	0.7
1	-0.4	1.0	0.4	-1.2	0.2	-0.5	0.6	-0.8	0.1	-0.3
2	1.1	-0.6	0.7	0.9	-0.3	0.1	-0.3	0.5	-0.6	0.9
3	-0.8	0.1	-0.5	1.3	0.5	-0.9	0.1	-0.2	0.4	-0.7
4	0.3	0.6	-1.1	-0.4	1.0	0.6	-0.7	0.3	-0.2	0.2
5	-0.1	-1.2	0.5	0.8	-0.2	1.0	0.9	-0.5	0.3	0.6
6	0.6	-0.7	0.9	0.2	-0.4	0.5	-0.6	0.7	-0.1	0.5
7	-1.0	0.4	-0.3	1.1	0.6	-0.2	0.2	-0.6	0.5	-0.4
8	0.2	-0.9	0.8	0.7	-0.1	0.4	-0.8	0.1	-0.7	0.6
9	-0.5	0.3	-0.7	1.0	0.3	-0.6	0.4	-0.3	0.2	-0.5
10	0.9	-0.4	1.2	0.3	-0.5	0.8	-0.2	0.6	-0.4	0.8
11	-0.6	0.8	0.1	-1.0	0.4	-0.3	0.7	-0.7	0.0	-0.2
12	0.4	-0.1	0.6	0.6	-0.2	0.2	-0.4	0.2	-0.3	0.4
13	-0.9	0.5	-0.4	1.2	0.7	-0.8	0.3	-0.4	0.6	-0.6
14	0.1	-0.8	0.3	0.9	-0.6	0.7	-0.5	0.3	-0.5	0.3
15	-0.2	0.7	0.2	-0.9	0.1	-0.4	0.8	-0.1	0.4	-0.1

Causal discovery aims to recover $\mathcal{G}$ from data.



Which arrows can data reveal?

3 Different approaches

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Three families of methods:

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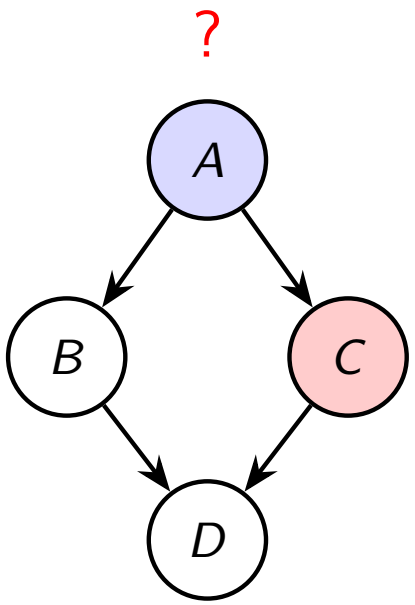
   Use conditional independence tests to rule out or confirm edges (PC algorithm).



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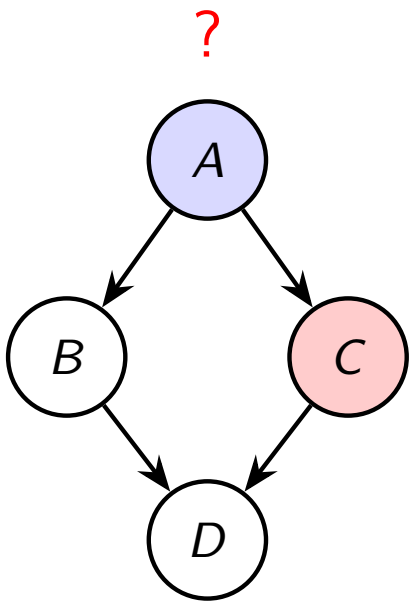

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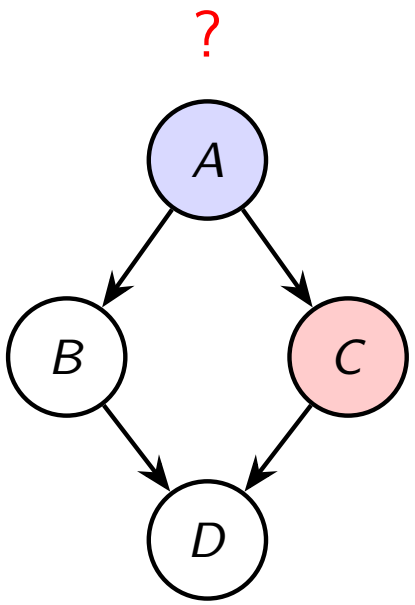
3. **Functional / asymmetry-based**
   Exploit the mathematical asymmetry between cause and effect (Additive Noise Models).



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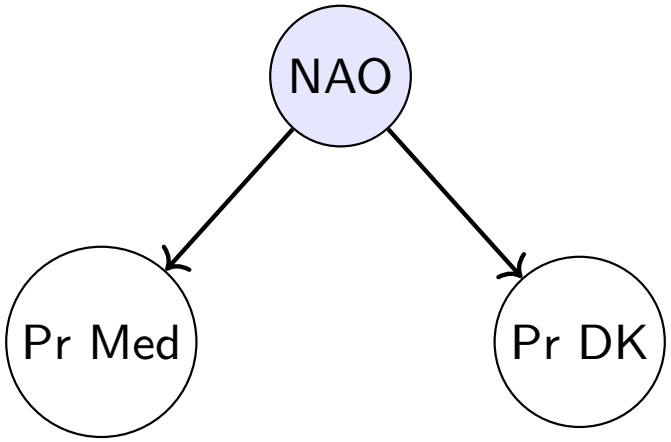
Goal: finding (d) separation in the graph to be able to recover it.

1. **Intuition**
   - $X \perp Y \mid Z$: Once we account for the state of Z, does X tell us anything new about Y?



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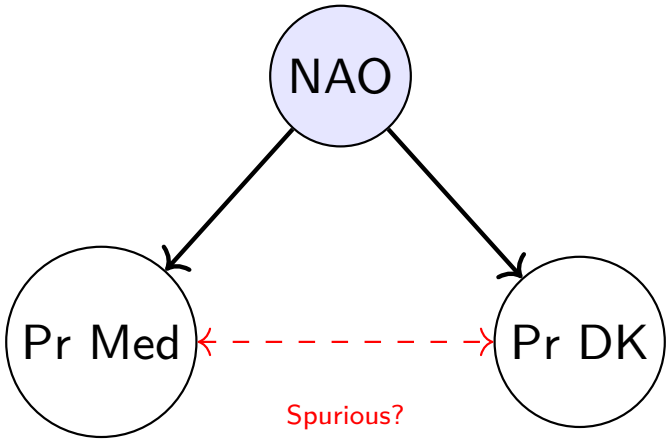

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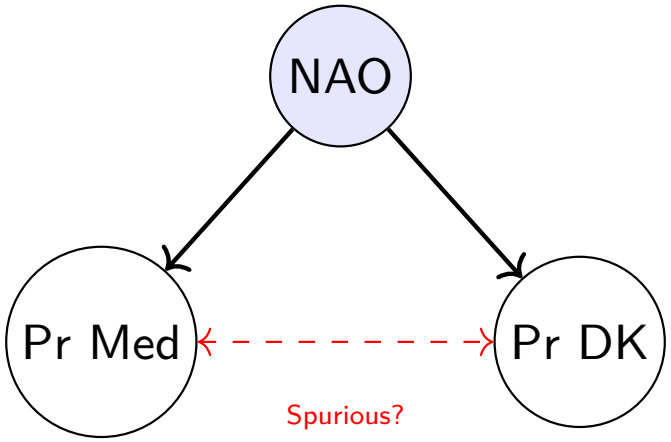
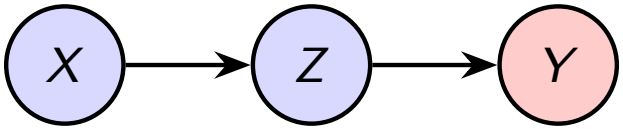
3. **Nonlinear CI**
   - **KCI:** Detects dependencies that linear tests miss.
   - Based on kernel dependence measure



Chain

$X \to Z \to Y$



Conditioning on Z **blocks** the path.

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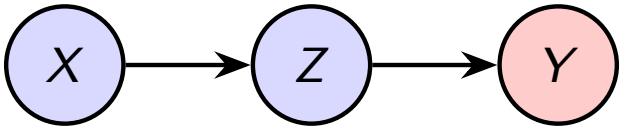
$X \rightarrow Z \rightarrow Y$



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Fork

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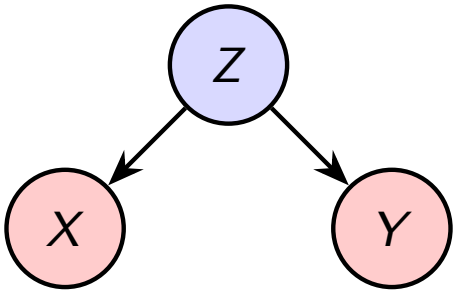

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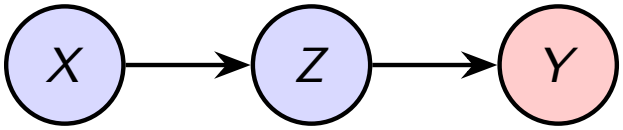

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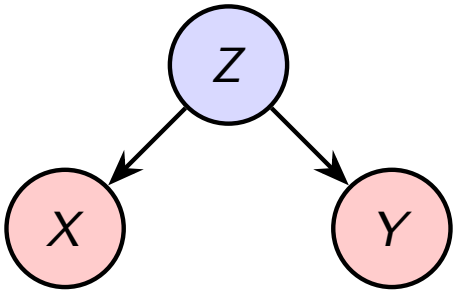

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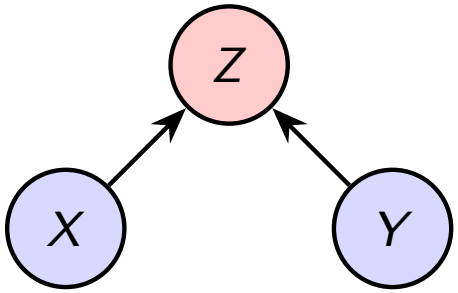
Collider

$X \rightarrow Z \leftarrow Y$



Conditioning on Z **opens** the path.

Fundamental limit: observational data[1] can only identify the **Markov Equivalence Class (MEC)** — the set of DAGs that encode the *same* conditional independencies.

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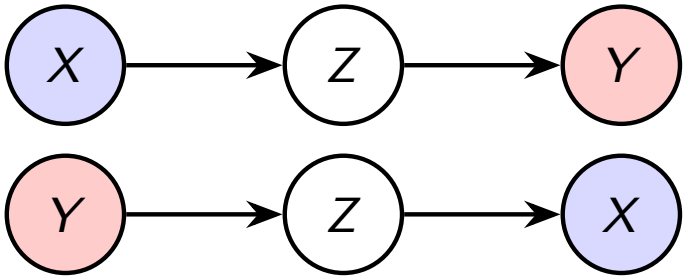

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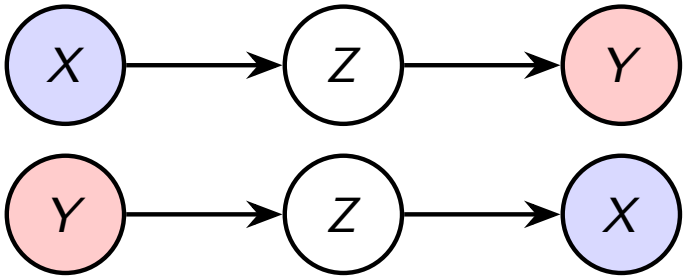

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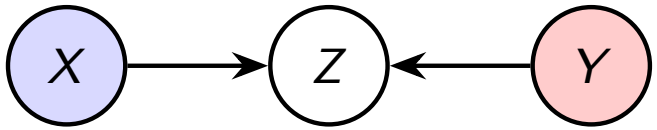

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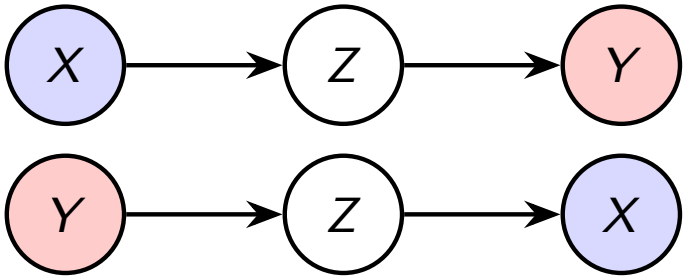

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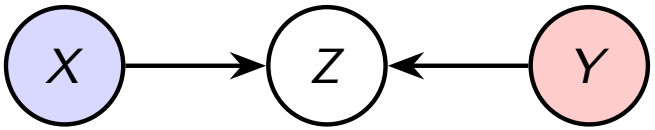
- The output is a **CPDAG** (Completed Partially Directed Acyclic Graph) — some edges remain undirected



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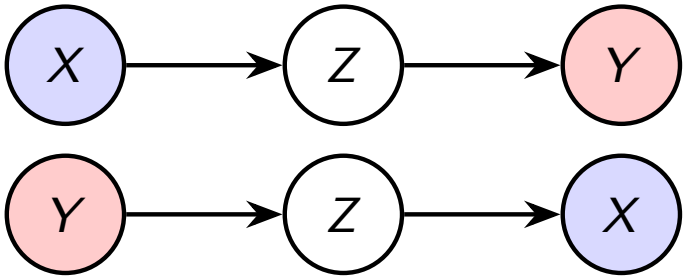

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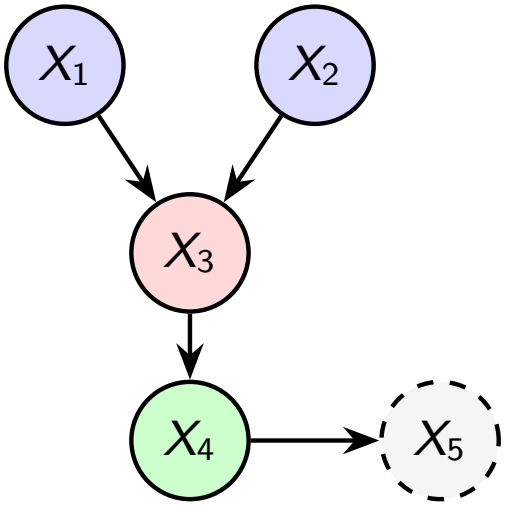
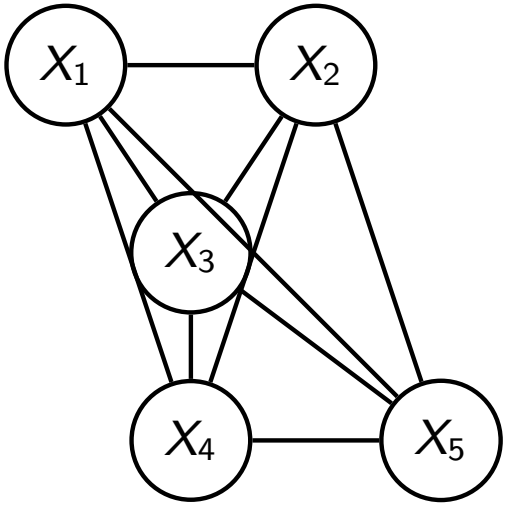
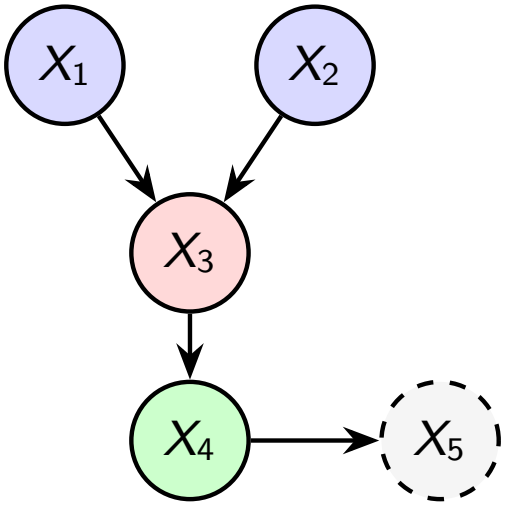


Step 1: Full graph





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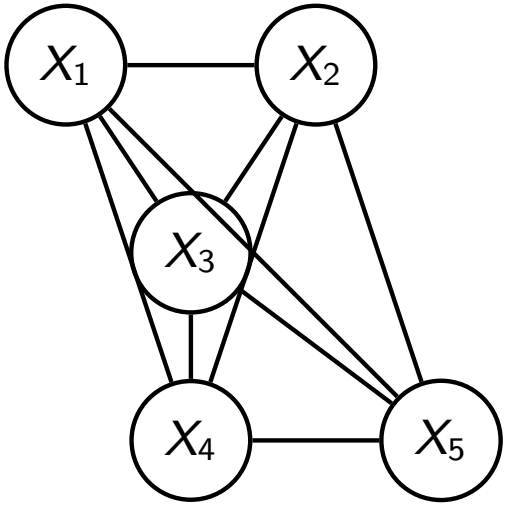
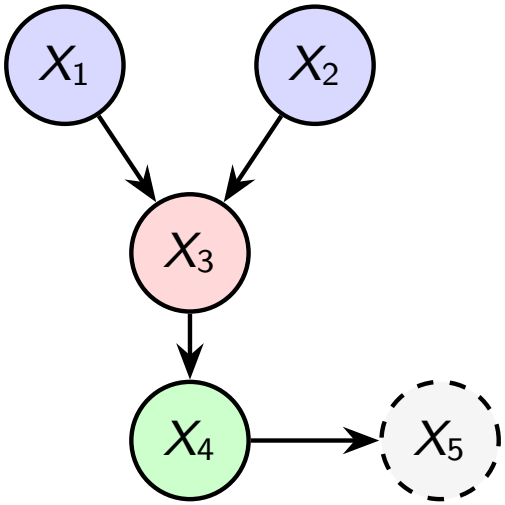


Test $X_i \perp X_j | S$



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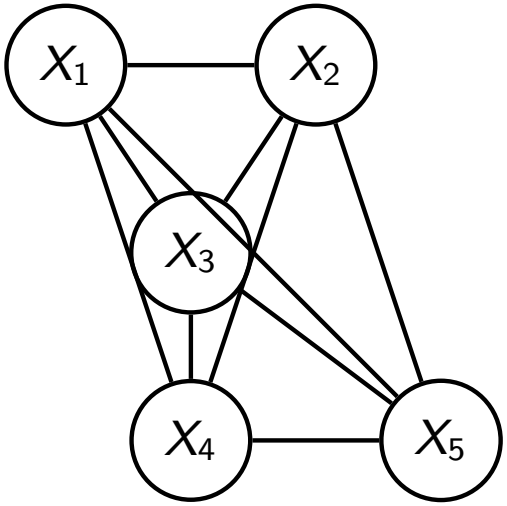

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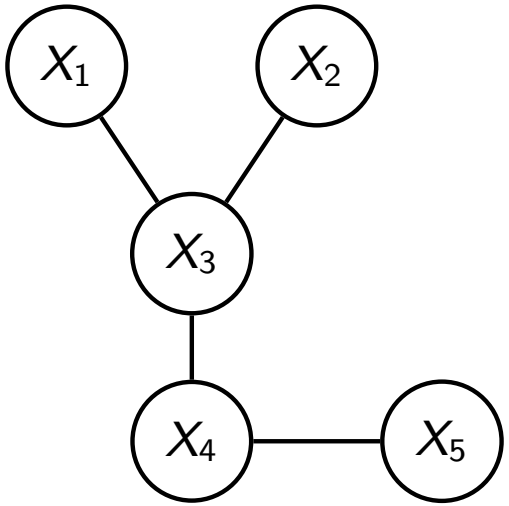
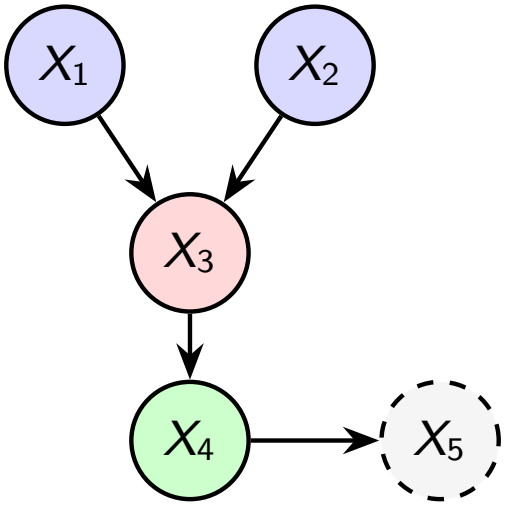
Step 2: Skeleton





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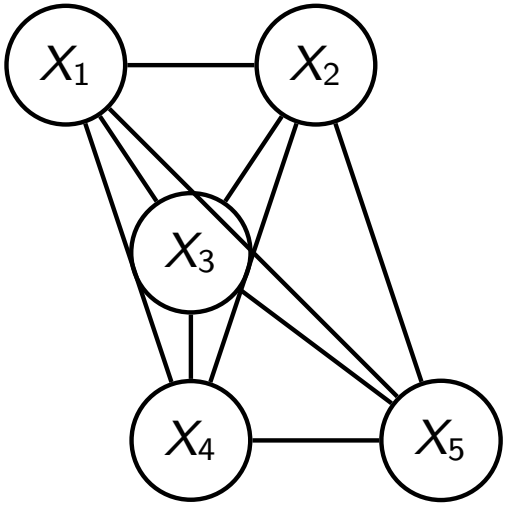

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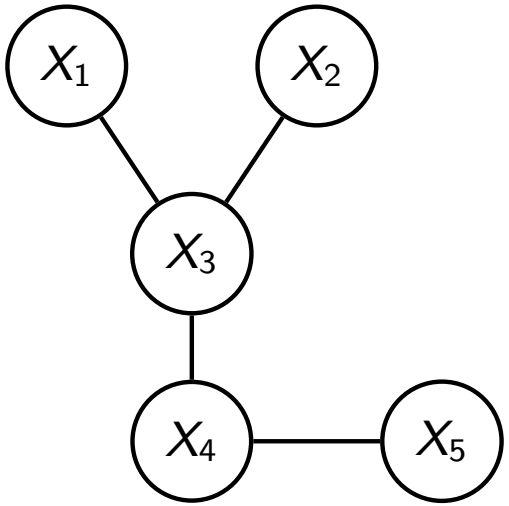

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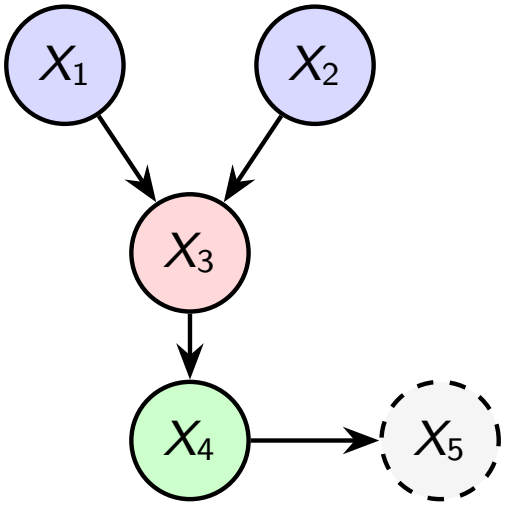

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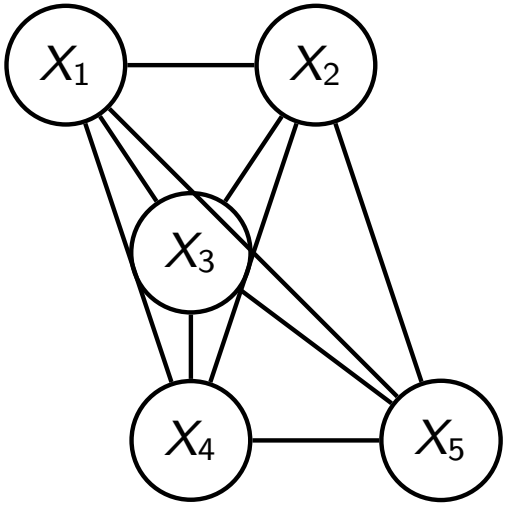

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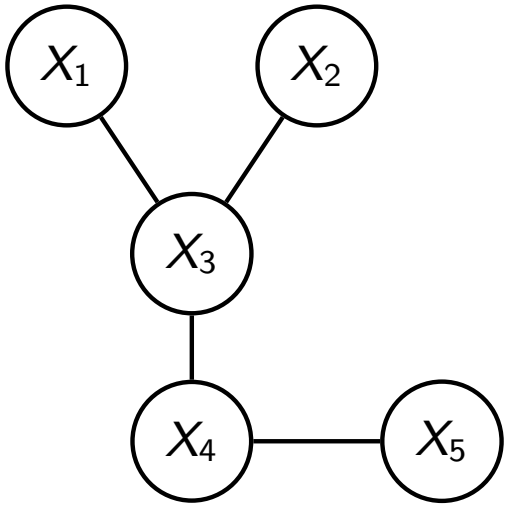

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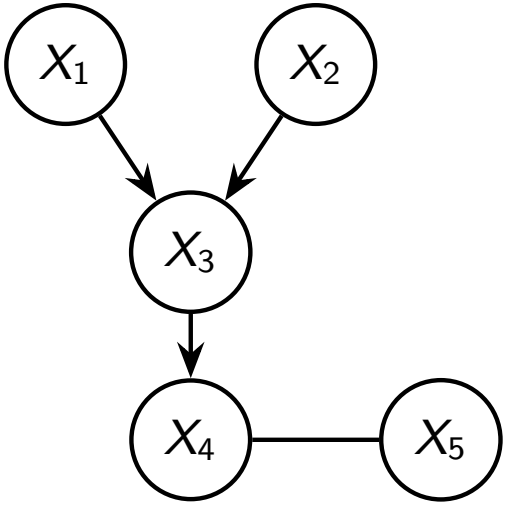

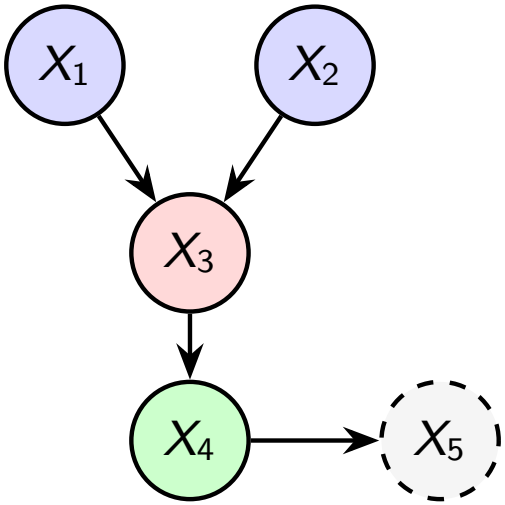

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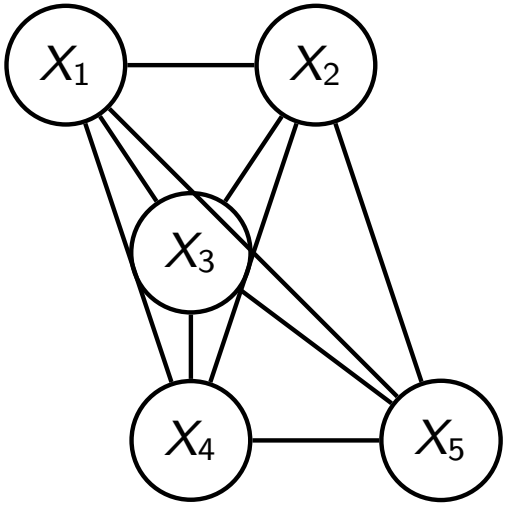

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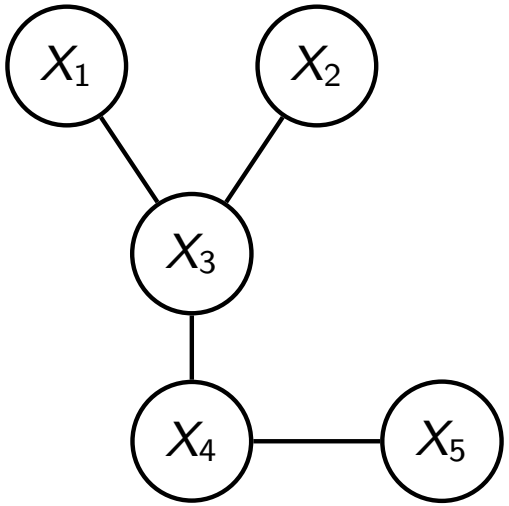

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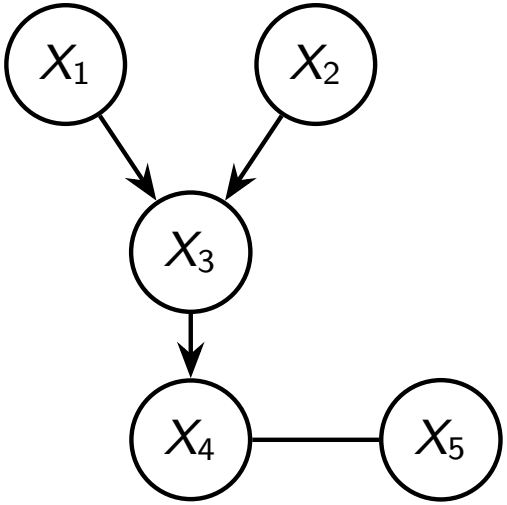

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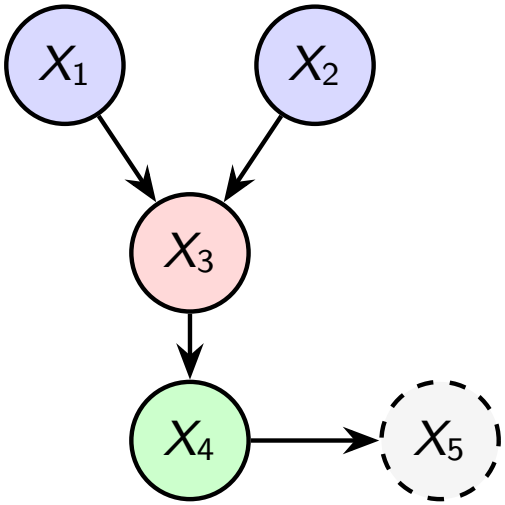

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- Same output as PC: a CPDAG (equivalence class).

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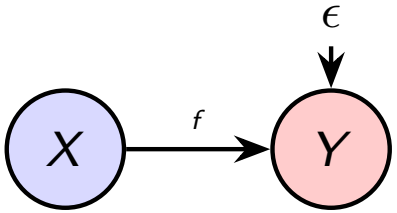

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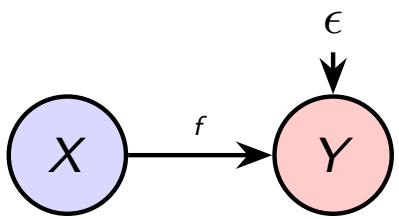

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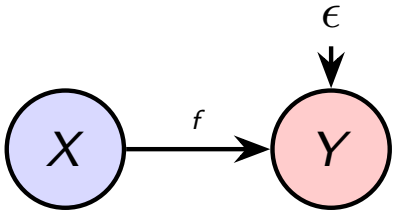

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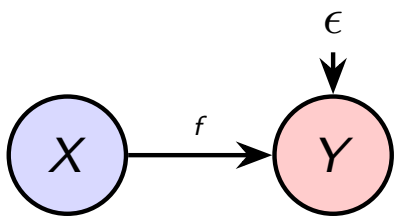

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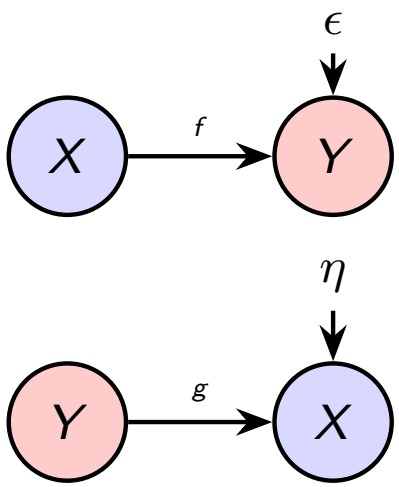

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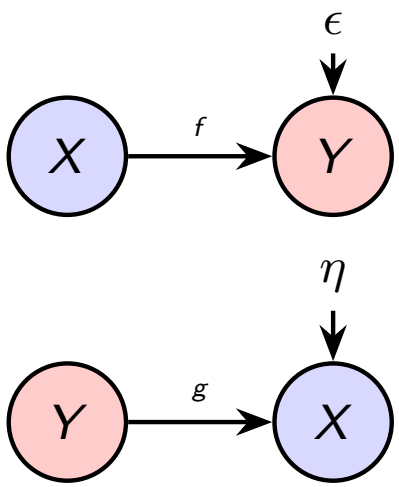

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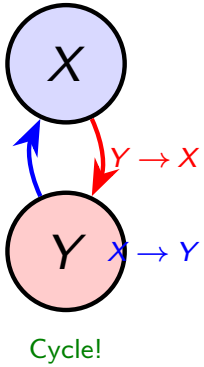

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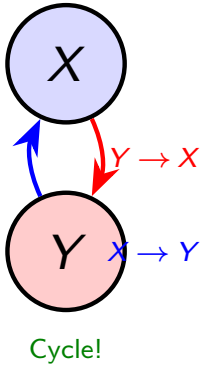

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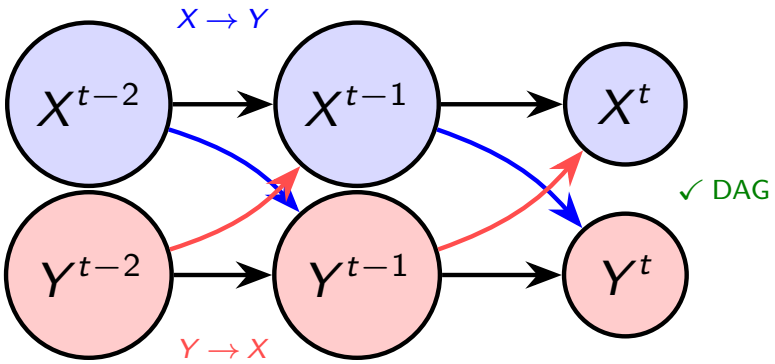
Time-unfolded DAG



Time-lagged nodes transform the instantaneous cycle into a DAG, allowing PC-style CI tests to work.

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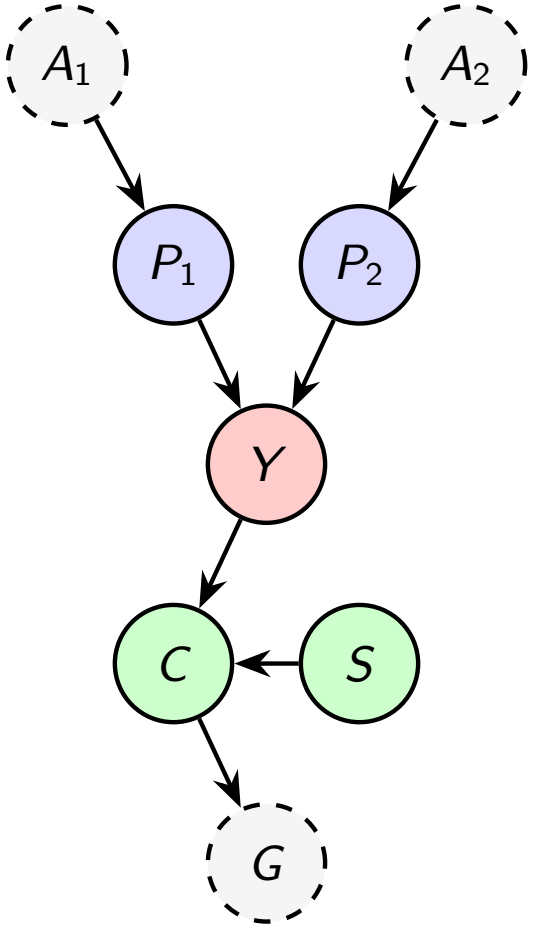

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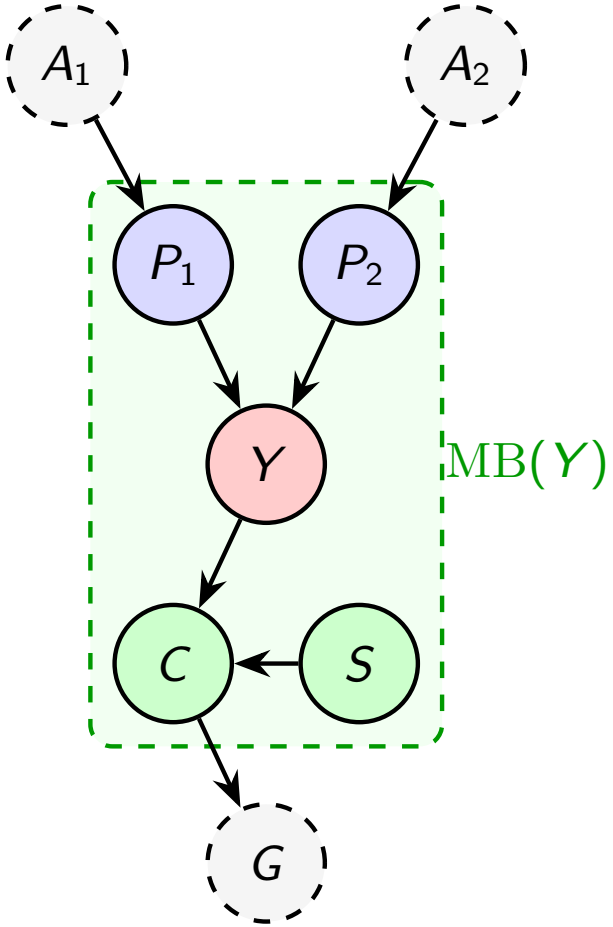

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Takeaway

Use **CI-based** or **score-based** methods when only MB is needed; use **Functional** methods when you need to distinguish *causes* from *effects* and *spouses*.

Causality-based Feature Selection: Methods and Evaluations

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Feature selection is a crucial preprocessing step in data analytics and machine learning. Classical feature selection algorithms select features based on the correlations between predictive features and the class variable and do not attempt to capture causal relationships between them. It has been shown that the knowledge about the causal relationships between features and the class variable has potential benefits for building interpretable and robust prediction models, since causal relationships imply the underlying mechanism of a system. Consequently, causality-based feature selection has gradually attracted greater attentions and many algorithms have been proposed. In this article, we present a comprehensive review of recent advances in causality-based feature selection. To facilitate the development of new algorithms in the research area and make it easy for the comparisons between new methods and existing ones, we develop the first open-source package, called CausalFS, which consists of most of the representative causality-based feature selection algorithms (available at https://github.com/kuiy/CausalFS). Using CausalFS, we conduct extensive experiments to compare the representative algorithms with both synthetic and real-world datasets. Finally, we discuss some challenging problems to be tackled in future research.

CCS Concepts: • **Computing methodologies → Feature selection**;

Additional Key Words and Phrases: Feature selection, Bayesian network, Markov boundary

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- Systematic description of the different algorithms and their general strategy

4 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

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$$Y^e = \beta^\top X_S^e + \epsilon^e$$

   if coefficients and residual distribution ϵ^e are stable across *all* environments e.
2. **Collect** all S that *pass* the invariance test.
3. **Output:** intersection of all passing sets

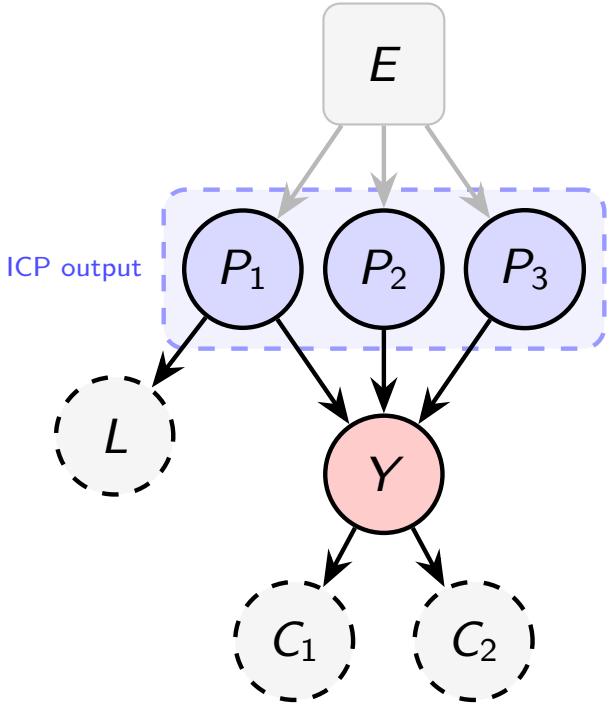
$$\widehat{\mathrm{Pa}(Y)} = \bigcap \{S : \text{invariant across all } e\}$$

Guarantee

With high probability, $\widehat{\mathrm{Pa}(Y)} \subseteq \mathrm{Pa}(Y)$ — no false positives.
ICP may miss some parents (low power), but the parents it finds are reliable.

Natural environments are already available

- Different **seasons** (DJF vs. JJA)



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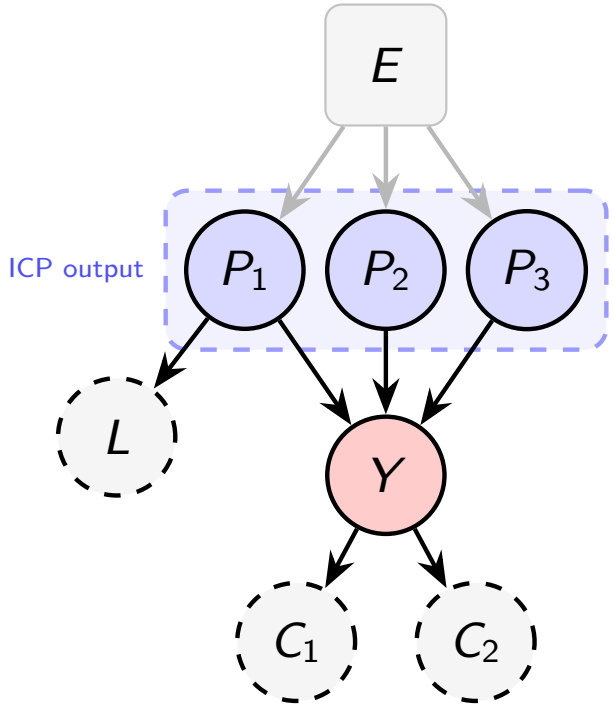

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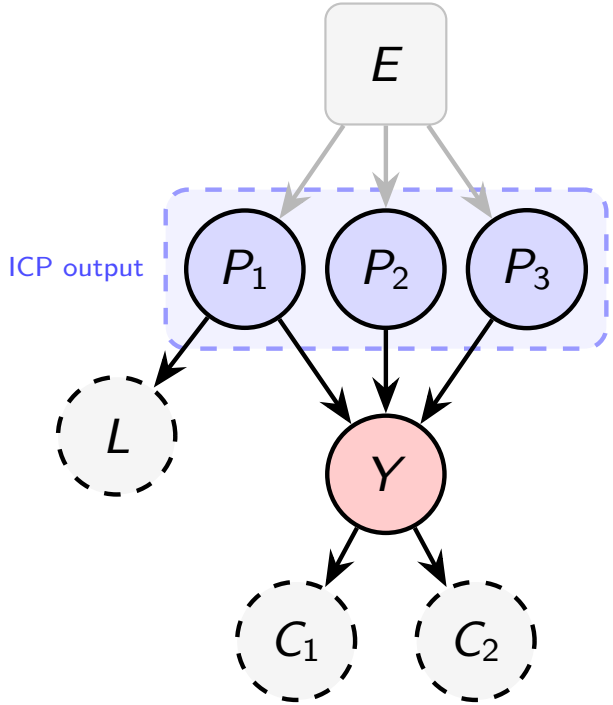

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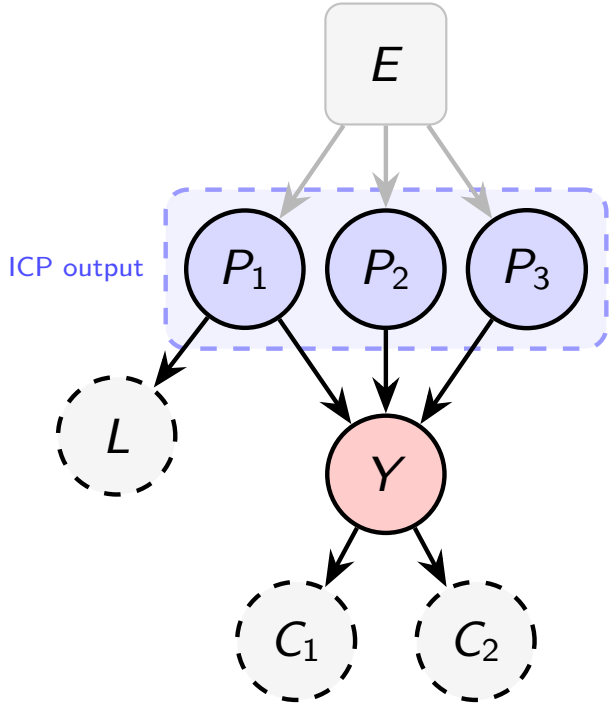

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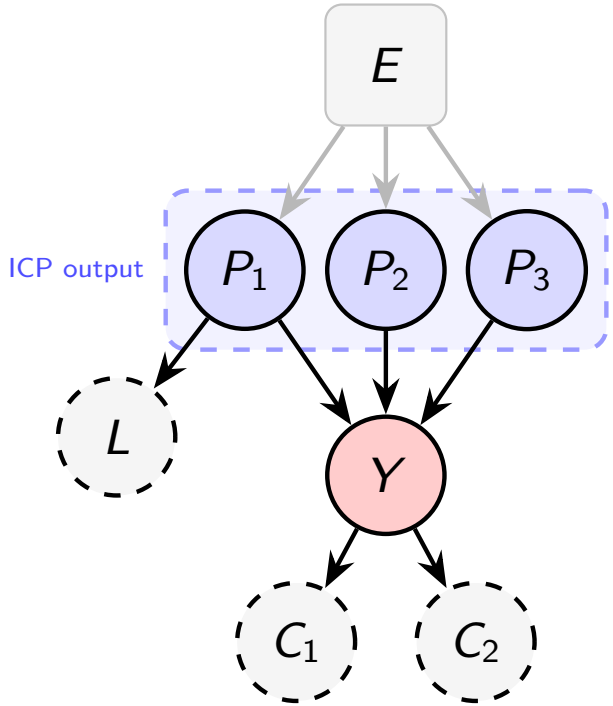

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- Very Similar to ICP but learns a **feature representation** instead of selecting features

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Instruments ("anchors") A affect X and Y. Anchor regression interpolates between OLS and Instrumental Variables (IV):

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The Core Insight

Enforcing invariance across environments implicitly selects causal features and discards spurious correlations — without requiring explicit causal graph recovery.

5 Outline

1. From ML to Robust ML
2. Robustness and Causality
3. Causal Discovery
4. Invariant Causal Prediction (ICP)
5. Summary and Discussion

5 Take-Aways

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Software: `causal-learn` (PC, GES, ANM) · `tigramite` (PCMCI) · `InvariantCausalPrediction` (ICP, R)